VOLATILITY FORECASTING PROJECT(IN-PROGRESS) Comparing Heterogeneous Autoregressive Model to Fractionally Integrated GARCH Model in predicting minute level Realized Volatility of Futures Data

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1 Introduction

This report summarizes an empirical study to determine whether a fractionally integrated GARCH (FI-GARCH) process can outperform a simple Ordinary Least Squares (OLS) based Heterogeneous AutoRegressive (HAR) model in forecasting out-of-sample volatility using 100ms high-frequency NQ futures data.

2 The Setup

Formally define the time series of the order book as $O = \{O_t, t \in [0, n] \cup \mathbb{N}\}$.
Let $O_t = \{BQ_t, AQ_t, BP_t, AP_t\}$.
Define the Micro Price (MP) as:

$$MP_t = \frac{BQ_t BP_t + AQ_t AP_t}{BQ_t + AQ_t}$$

Using the micro price, which structurally resembles a volume-weighted average price, we compute the financial returns. Treating the micro price as a robust proxy for the actual asset price, the log-return series $r = \{r_t\}$ is defined as:

$$r_t = \ln(MP_t) - \ln(MP_{t-1}), \quad t > 0 \tag{1}$$

At this ultra-high-frequency scale, the numerical difference between simple returns and log-returns is marginal; however, log-returns are strictly preferred in time-series analysis to induce stationarity.

To forecast volatility, Realized Volatility (RV) is utilized as a non-parametric proxy. The 100ms data is resampled to 1-minute and 5-minute intervals to compute the realized volatility, which serve as the primary target variables for forecasting. To compare models across differing frequencies, 1-minute forecasts can be aggregated to a 5-minute frequency by summing the predicted variances within each 5-minute interval.

Realized Volatility over interval d is formally defined as:

$$X_t^d = \sum_{\tau=t-d}^{t-1} r_\tau^2 \tag{2}$$

The structural form of the HAR model is specified as:

$$X_t = \beta_0 + \beta_s X_{t-1}^s + \beta_m X_{t-1}^m + \beta_l X_{t-1}^l \tag{3}$$

where X_t is the realized volatility, and s, m, and l represent the short-term, medium-term, and long-term aggregation horizons respectively ($s < m < l$, with $s, m, l \in \mathbb{N}$).

The HAR model relies on standard OLS to estimate the coefficients, minimizing the standard quadratic loss function:

$$L = \sum_{t \in \mathcal{T}} (X_t - \hat{X}_t)^2$$

where $\mathcal{T}$ represents the overall test set.

Conversely, the FIGARCH model is defined by the conditional variance equation:

$$X_t^2 = \omega^* + \left[1 - \frac{\phi(L)}{\beta(L)}(1-L)^{d_v}\right]\epsilon_t^2 \tag{4}$$

Where:

- X_t^2 is the conditional variance.
- ω^* is the adjusted baseline variance constant.
- d_v is the fractional integration parameter ($0 \leq d_v \leq 1$).
- $\phi(L)$ and $\beta(L)$ are the ARCH and GARCH polynomials in the lag operator.
- ϵ_t^2 represents the squared residuals extracted from the mean equation.

The parameters for FIGARCH are estimated via Maximum Likelihood Estimation (MLE), minimizing the negative log-likelihood function:

$$\text{NLL}(\theta) = -\sum_{t \in \mathcal{T}} \log P(X_t \mid p, q, d_v)$$

Data

The underlying dataset consists of NQ futures 100ms limit order book data spanning from November 3, 2025, to November 11, 2025.

Implementation

For the HAR model estimation, the **statsmodels** API in Python is utilized. For the FIGARCH model, **ARFIMA**, a library for fractional processes, is used.

Rolling Window Forecast Framework

Let $Tr_t = \{X_k, k \in [t, t + w_{Tr}] \cap \mathbb{N}\}$ be the training window of size w_{Tr}, and $Ts_t = \{X_k, k \in (t + w_{Tr}, t + w_{Tr} + w_{Ts}] \cap \mathbb{N}\}$ be the immediate out-of-sample testing window of size w_{Ts}.

The models are fitted dynamically using the training window to estimate the optimal parameters, which are then used to predict the immediate testing window.

3 Results

The out-of-sample performance is evaluated using Mean Squared Error (MSE), the asymmetric QLIKE loss function, Heteroscedastic MSE (HMSE), and the Mincer-Zarnowitz R^2.

3.1 Metrics Definitions

- **Mean Squared Error (MSE):**

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{X}_i)^2 \tag{5}$$

- **Average QLIKE Loss:**

$$QLIKE = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_i}{\hat{X}_i} - \ln\left(\frac{X_i}{\hat{X}_i}\right) - 1\right) \tag{6}$$

- **Heteroscedastic MSE (HMSE):**

$$HMSE = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{X_t}{\hat{X}_t} - 1\right)^2 \tag{7}$$

- **Mincer-Zarnowitz R^2 (MZ-R^2):**
  First, the observed variance is regressed on the forecasted variance:

$$X_t = \alpha + \beta\hat{X}_t + \epsilon_t \tag{8}$$

  The MZ-R^2 is then computed as:

$$MZ\text{-}R^2 = 1 - \frac{\sum_{t=1}^{n}(X_t - \hat{\alpha} - \hat{\beta}\hat{X}_t)^2}{\sum_{t=1}^{n}(X_t - \overline{X})^2} \tag{9}$$

Table 1 summarizes the comparative performance across the evaluated models and sampling frequencies.

Table 1: Out-of-Sample Forecasting Metrics

Model Framework	**Frequency**	**MSE**	**QLIKE**	**HMSE**	**MZ R^2**
HAR-RV	1-Minute	6.36×10^{-15}	1.6648	1494.34	0.8188
HAR-RV	5-Minute	1.46×10^{-13}	1.3099	966.15	0.7869
FIGARCH	1-Minute	1.19×10^{-12}	1.1944	0.58	0.0392
FIGARCH	5-Minute	2.45×10^{-12}	1.2868	12.19	0.1937

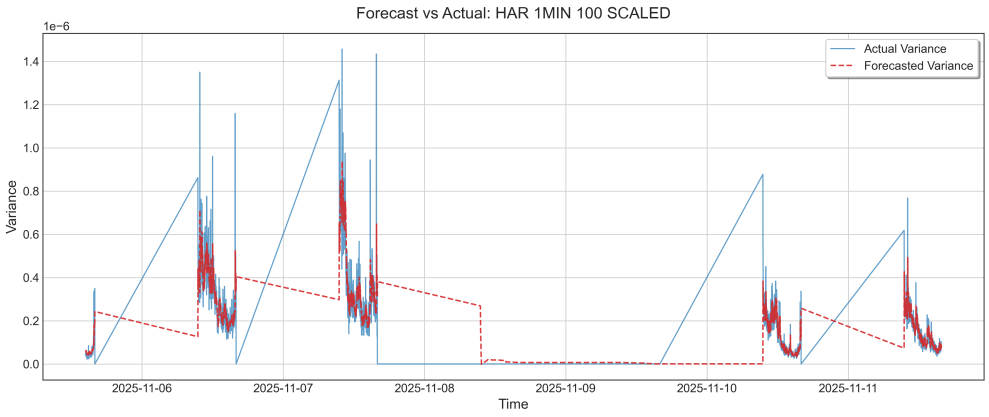


Figure 1: Intraday Realized Volatility vs. HAR Forecast (1 min)

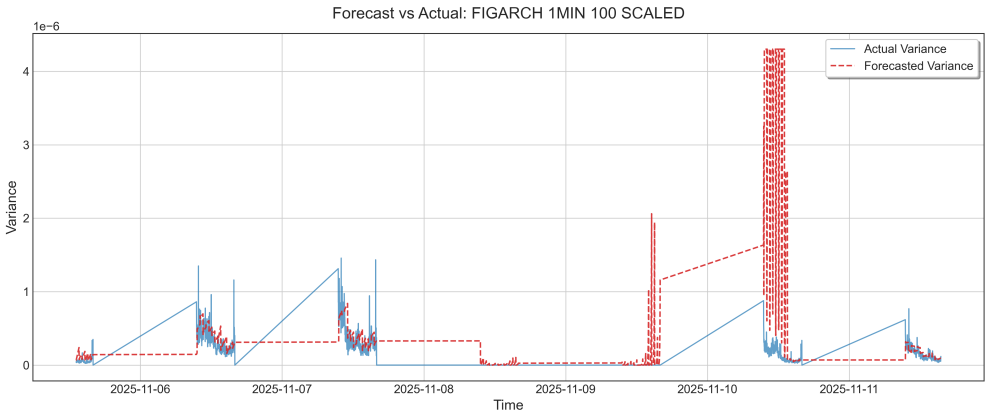


Figure 2: Intraday Realized Volatility vs. FIGARCH Forecast (1 min)

3.2 Diebold-Mariano Test Results

To systematically assess the statistical significance of the discrepancies in predictive accuracy, a one-sided Diebold-Mariano (DM) test was conducted on the aligned out-of-sample forecasts.

Let $d(t) = \text{Loss}_1(t) - \text{Loss}_2(t)$ denote the loss differential at time t between model M_1 and model M_2. The null hypothesis posits that M_1 is strictly less than or equal to M_2 in predictive accuracy, tested against the one-sided alternative that M_1 is strictly superior.

$$H_0 : \mathbb{E}[d(t)] \geq 0 \tag{10}$$

$$H_1 : \mathbb{E}[d(t)] < 0 \tag{11}$$

To ensure time parity, models operating at a 1-minute frequency were temporally aggregated to match the 5-minute sampling frequency where applicable.

Cross-Model Evaluations:

- $M_1 = \text{HAR (5-min)}, \quad M_2 = \text{HAR (1-min aggregated)}$
  $n = 430, \quad t_{stat} = 3.39, \quad p = 0.99$
- $M_1 = \text{FIGARCH (5-min)}, \quad M_2 = \text{HAR (1-min aggregated)}$
  $n = 491, \quad t_{stat} = 5.59, \quad p = 0.99$
- $M_1 = \text{FIGARCH (1-min)}, \quad M_2 = \text{HAR (1-min)}$
  $n = 2412, \quad t_{stat} = 14.27, \quad p = 0.99$

Conclusion: In all three scenarios, the calculated test statistics are massively positive, yielding p-values of approximately 0.99. Consequently, we fail to reject the null hypothesis in the specified direction. These overwhelmingly positive t-statistics mathematically prove that the loss differential $\mathbb{E}[d(t)]$ is highly positive, meaning the error of M_1 is consistently larger than the error of M_2.

Ultimately, the 1-minute HAR model definitively outperformed the 5-minute HAR, the 5-minute FIGARCH, and the 1-minute FIGARCH models. This firmly suggests that despite the theoretical sophistication of fractional integration, the simple OLS-based HAR architecture is vastly superior for extracting the true long-memory dynamics of ultra-high-frequency realized volatility.